

Sude Ozkaya

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EDUCATION

Carnegie Mellon University, Pittsburgh, PA

May 2028

B.S. in Cognitive Science and Computer Science, Minor in Creative Writing

Cumulative GPA: 3.8/4.0 | Dietrich College Dean's List (Spring 2025, Fall 2025, Spring 2026)

Relevant Coursework: Data Structures & Algorithms, Functional Programming, Discrete Mathematics, Linear Algebra, Statistics & Data Science, Designing Human-Centered Software, Persuasive Design

EXPERIENCE

NSF AI Institute for Artificial and Natural Intelligence REU, Columbia University

May 2026 – August 2026

Machine Learning Research Intern; P.I. Prof. Nikolaus Kriegeskorte, mentored by Dr. Akihito Maruya

- Extended an existing **PyTorch**-based computer vision and video-synthesis pipeline to generate controlled synthetic motion-texture stimuli for perceptual experiments.
- Modified spatiotemporal feature representations and optimization procedures to manipulate higher-order motion statistics while controlling lower-level visual features across space, time, orientation, and scale.
- Built a reproducible synthetic-data generation and evaluation workflow, including graded naturalness conditions and a **3AFC behavioral experiment** to measure perceptual sensitivity to motion statistics.

Social World Understanding, Carnegie Mellon University, Robotics Institute

January 2026 – September 2026

AI Research Assistant; P.I. Prof. Reid Simmons, mentored by Dr. Pragathi Praveena

- Developed a story-based **QA system** for reasoning about social relationships, beliefs, and intentions
- Prototyped information-retrieval and structured narrative-parsing pipelines to extract relevant social context from natural-language stories
- Designed a query taxonomy and evaluation framework to benchmark QA performance across categories of social reasoning tasks

Furhat Robotics Emotion Mimic Project, Koc University

May 2025 – August 2025

AI & Robotics Research Intern; P.I. Prof. Engin Erzin

- Developed an **LLM/NLP-powered** human-robot interaction system generating emotion-adaptive responses from natural-language input.
- Integrated an **LLM, FastAPI, and the Furhat SDK** for real-time expressive speech, facial behavior, and emotion scoring.

Stanford SHTM Program, Stanford University

June 2023 – August 2023

Machine Learning Research Intern; P.I. Prof. Mert Pilanci

- Developed a neural audio-generation approach using adaptive crossfading and built a preprocessing pipeline with **Spleeter, Librosa, and NumPy** for the 150-song MUSDB18 dataset.
- Presented at the Stanford SHTM Symposium and co-authored “Bridging the Gap in Generating A.I. for Audio Generation”, published in *The Informaticists*

LEADERSHIP & PROJECT EXPERIENCE

Girls Who Code, CMU Chapter, *President*

December 2025 – Present

- Lead a 36 chapter serving 36 middle school students across 3 programs, coordinating 25+ student instructors and semester-long Python and AI curricula.
- Manage weekly executive and chapter meetings, technical curriculum, guest speakers, and program operations; and teach Python while mentoring instructors in project-based CS education

Gesture-Controlled Cut the Rope Game

April 2025

- Built a computer-vision-controlled game using **Python, CMU Graphics, OpenCV, and MediaPipe**, translating real-time hand tracking into gameplay controls.

English with Fun and Games, Robert College *Project Leader*

September 2022 – March 2023

- Led an **18-member volunteer** team developing an elementary-school English education program

SKILLS

Languages: Python, C, Java, SML, R, HTML/CSS

Frameworks & Tools: PyTorch, LLMs, OpenCV, MediaPipe, PsychoPy, FastAPI, Figma, Librosa, Spleeter, NumPy

ACTIVITIES

The Tartan, CMU Student Newspaper, Staffwriter (2025 – Present) | **Women's Basketball Club**, CMU, Robert College, Aci Schools (2012 – Present)